INCH-POUND

MIL-DTL-27426/2
25 SEPTEMBER 1997

DETAIL SPECIFICATION

RINGS, RETAINING, SPIRAL
(UNIFORM CROSS SECTION)
HEAVY DUTY EXTERNAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-27426, listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

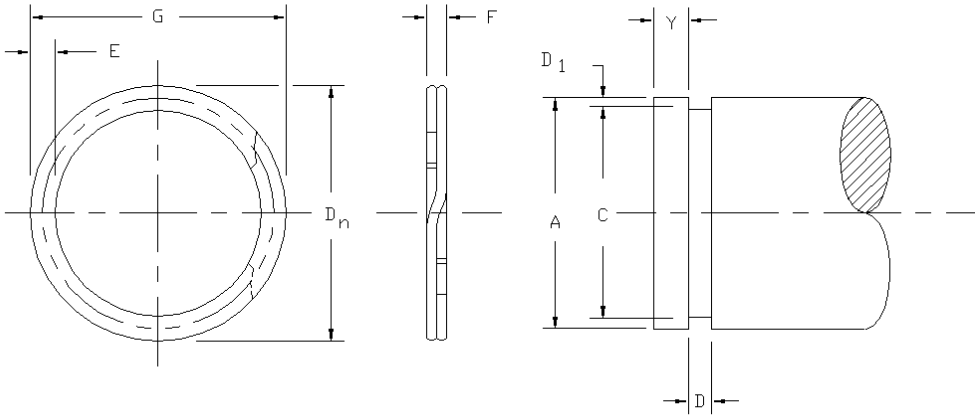


FIGURE 1. EXTERNAL RING

Maximum bottom groove radius is 0.005 inches for rings up to 0.946 inches free diameter and 0.010 inches for larger rings. Ring thickness applies only to unplated rings. Add 0.002 inches upper thickness tolerance for plated rings. Groove widths listed will accept rings without modification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : the Defense Industrial Supply Center, (Code DISC-EED), 700 Robbins Avenue, Philadelphia, PA 19111-5096, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. HEAVY DUTY EXTERNAL RETAINING RINGS.

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0100	.469	11.9	.439		.025		.045		.443		.029		.013
0101	.500	12.7	.464		.035		.050		.468		.039		.016
0102	.551	14.0	.514		.035		.050		.519	±.002	.039		.016
0103	.562	14.3	.525		.035		.050		.530		.039		.016
0104	.594	15.1	.554		.035		.050		.559		.039		.017
0105	.625	15.9	.583		.035		.055		.588		.039		.018
0106	.669	17.0	.623		.035		.055		.629		.039	+.003	.020
0107	.688	17.5	.641		.042		.065		.646		.046	−.000	.021
0108	.750	19.0	.698		.042		.065		.704		.046		.023
0109	.781	19.8	.727		.042		.065		.733	±.003	.046		.024
0110	.812	20.6	.756	+.000	.042		.065		.762		.046		.025
0111	.875	22.2	.814	−.013	.042	±.002	.075		.821		.046		.027
0112	.938	23.8	.875		.042		.075		.882		.046		.028
0113	.984	25.0	.919		.042		.085		.926		.046		.029
0114	1.000	25.4	.932		.042		.085		.940		.046		.030
0115	1.023	26.0	.953		.042		.085	+.003	.961		.046		.031
0116	1.062	27.0	.986		.050		.103	−.005	.998		.056		.032
0117	1.125	28.6	1.047		.050		.103		1.059		.056		.033
0118	1.188	30.2	1.105		.050		.103		1.118		.056		.035
0119	1.250	31.7	1.163		.050		.103		1.176	±.004	.056		.037
0120	1.312	33.3	1.218		.050		.118		1.232		.056		.040
0121	1.375	34.9	1.277		.050		.118		1.291		.056		.042
0122	1.438	36.5	1.336		.050		.118		1.350		.056		.044
0123	1.500	38.1	1.385		.050		.118		1.406		.056	+.004	.047
0124	1.562	39.6	1.453		.062		.128		1.468		.068	−.000	.047
0125	1.625	41.7	1.513		.062		.128		1.529		.068		.048
0126	1.687	42.8	1.573		.062		.128		1.589		.068		.049
0127	1.750	44.4	1.633		.062		.128		1.650	±.005	.068		.050
0128	1.771	44.9	1.651	+.000	.062		.128		1.669		.068		.051
0129	1.812	46.0	1.690	−.020	.062		.158		1.708		.068		.052
0130	1.875	47.6	1.751		.062		.158		1.769		.068		.053
0131	1.969	50.0	1.838		.062		.158		1.857		.068		.056
0132	2.000	50.8	1.867		.062		.158		1.886		.068		.057
0133	2.062	52.3	1.932		.078		.168		1.946		.086		.058
0134	2.125	53.9	1.989		.078		.168		2.003		.086		.061
0135	2.156	54.7	2.018		.078	+.003	.168		2.032		.086		.062
0136	2.250	57.1	2.105		.078		.168		2.120		.086		.065
0137	2.312	58.7	2.163		.078		.168		2.178		.086		.067
0138	2.375	60.3	2.223	+.000	.078		.200	+.004	2.239		.086		.068
0139	2.437	61.9	2.283	−.025	.078		.200	−.006	2.299		.086		.069
0140	2.500	63.5	2.343		.078		.200		2.360		.086	+.005	.070
0141	2.559	64.9	2.402		.078		.200		2.419		.086	−.000	.071
0142	2.625	66.6	2.464		.078		.200		2.481	±.006	.086		.072
0143	2.687	68.2	2.523		.078		.200		2.541		.086		.073
0144	2.750	69.8	2.584		.093		.225		2.602		.103		.074
0145	2.875	73.0	2.702		.093		.225		2.721		.103		.077
0146	2.937	74.5	2.760		.093		.225		2.779		.103		.079
0147	3.000	76.1	2.818		.093		.225		2.838		.103		.081
0148	3.062	77.7	2.878		.093		.225		2.898		.103		.082
0149	3.125	79.3	2.936		.093		.225		2.957		.103		.084
0150	3.156	80.1	2.965	+.000	.093		.225		2.986		.103		.085
0151	3.250	82.5	3.054	−.030	.093		.225		3.076		.103		.087
0152	3.344	84.9	3.144		.093		.225		3.166		.103		.090
0153	3.437	87.2	3.234	+.000	.093		.225		3.257		.103		.090

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TABLE I. MEDIUM DUTY EXTERNAL RETAINING RINGS. (CONT'D)

DASH NO.	SHAFT DIAMETER		RING DIMENSIONS						GROOVE DIMENSIONS				
	INCHES A	MM	FREE DIAMETER G	TOL	THICKNESS F	TOL	RADIAL WALL E	TOL	DIAMETER C	TOL	WIDTH D	TOL	NOMINAL GROOVE DEPTH D_1
0154	3.500	88.8	3.293	−.030	.111		.270		3.316		.120		.092
0155	3.543	89.9	3.333		.111		.270	+.004	3.357		.120		.093
0156	3.625	92.0	3.411		.111		.270	−.006	3.435		.120		.099
0157	3.687	93.6	3.469		.111		.270		3.493		.120		.097
0158	3.750	95.2	3.527	+.000	.111		.270		3.552	±.006	.120	+.005	.099
0159	3.875	98.3	3.647	−.035	.111	+.003	.270		3.673		.120	−.000	.101
0160	3.938	99.9	3.708		.111		.270		3.734		.120		.102
0161	4.000	101.5	3.765		.111		.270		3.792		.120		.104
0162	4.250	107.9	4.037		.111		.270		4.065		.120		.092
0163	4.375	110.0	4.161		.111		.270		4.190		.120		.092
0164	4.500	114.2	4.280		.111		.270		4.310		.120		.095
0165	4.750	120.6	4.518		.111		.270		4.550		.120		.100
0166	5.000	126.9	4.756		.111		.270		4.790		.120		.105
0167	5.250	133.2	4.995		.127		.350		5.030		.139		.110
0168	5.500	140.0	5.228	+.000	.127		.350		5.265		.139	+.006	.117
0169	5.750	145.9	5.466	−.050	.127	±.004	.350		5.505	±.007	.139	−.000	.122
0170	6.000	152.3	5.705		.127		.350	+.004	5.745		.139		.127
0171	6.250	158.6	5.938		.156		.418	−.008	5.985		.174		.132
0172	6.500	165.0	6.181	+.000	.156		.418		6.225		.174		.137
0173	6.750	171.3	6.410	−.060	.156		.418		6.465		.174		.142
0174	7.000	177.7	6.648		.156		.418		6.705		.174		.147
0175	7.250	184.0	6.891		.156		.437		6.942		.174		.154
0176	7.500	190.4	7.130		.187		.437		7.180		.209		.160
0177	7.750	196.8	7.368		.187		.437		7.420		.209		.165
0178	8.000	203.0	7.606		.187		.437		7.660	±.008	.209		.170
0179	8.250	209.8	7.845	+.000	.187		.437	+.004	7.900		.209		.175
0180	8.500	215.7	8.083	−.070	.187		.437	−.008	8.140		.209		.180
0181	8.750	222.2	8.324		.187		.437		8.383		.209		.182
0182	9.000	228.4	8.560		.187		.500		8.620		.209		.190
0183	9.250	235.0	8.798		.187		.500		8.860		.209		.195
0184	9.500	241.1	9.036		.187		.500		9.100		.209		.200
0185	9.750	248.0	9.275		.187		.500		9.338		.209	+.008	.206
0186	10.000	253.8	9.508		.187		.500		9.575		.209	−.000	.212
0187	10.250	260.4	9.754		.187	±.005	.500		9.814		.209		.218
0188	10.500	266.7	9.984		.187		.500		10.054		.209		.223
0189	10.750	273.1	10.221		.187		.500		10.293		.209		.228
0190	11.000	279.4	10.459		.187		.500		10.533		.209		.238
0191	11.250	285.8	10.692	+.000	.187		.500		10.772	±.010	.209		.239
0192	11.500	292.1	10.934	−.080	.187		.562		11.011		.209		.244
0193	11.750	298.5	11.171		.187		.562		11.250		.209		.250
0194	12.000	304.8	11.410		.187		.562		11.490		.209		.255
0195	12.250	311.2	11.647		.187		.562		11.729		.209		.260
0196	12.500	317.5	11.885		.187		.562		11.969		.209		.265
0197	12.750	323.9	12.124		.187		.562		12.208		.209		.271
0198	13.000	330.2	12.361		.187		.662		12.448		.209		.276
0199	13.250	336.6	12.598		.187		.662		12.687		.209		.281
0200	13.500	342.9	12.837	+.000	.187		.662		12.927		.209		.286
0201	13.750	349.3	13.074	−.090	.187		.662		13.166	±.012	.209		.292
0202	14.000	355.6	13.311		.187		.662		13.405		.209		.297
0203	14.250	362.0	13.548		.187		.662		13.644		.209		.303
0204	14.500	368.3	13.787		.187		.750		13.884		.209		.308
0205	14.750	374.7	14.024		.187		.750		14.123		.209		.313
0206	15.000	381.0	14.262		.187		.750		14.363		.209		.318

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